

When Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection:

A Bilingual Diagnosis of Class-Conditional Negative Transfer

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Abstract

Sarcasm is non-literal language whose intended meaning may diverge from apparent sentiment, complicating detection in online text. As LLM-generated content appears alongside human writing, systems may need to judge both sarcasm and machine authorship. These judgments also interact in generation, where detector scores may be treated as measures of sarcastic intent and human authorship. Although sarcasm detection and machine-generated text detection (MGTD) are studied separately, their interaction in shared representations remains insufficiently understood. MGTD may promote source-predictive stylistic or distributional cues, whereas sarcasm detection should reflect pragmatic intent rather than source. We conduct, to our knowledge, the first bilingual diagnosis of how MGTD supervision affects sarcasm detection across Chinese and English. We harmonize bilingual resources, generate topic-matched AI texts, and build DualTest with source and sarcasm labels. Architecturally matched joint and single-task models reveal class-conditional negative transfer concentrated on AI-generated text prompted to be non-sarcastic. This pattern persists with a second generator and is attenuated by RCNN and interaction-projection modules. An exploratory comparison shows that detector scores for human authorship and LLM ratings of perceived human-likeness rank generated outputs differently. These findings motivate subgroup evaluation and caution against detector-only assessment.

1 Introduction

A sarcastic sentence typically conveys an attitude or intent that differs from, and may oppose, its literal meaning. Sarcasm can invert sentiment polarity and disrupt opinion-mining systems (Tay et al., 2018); ironic or sarcastic abuse is also a known challenge for content moderation (Vidgen et al., 2019). The task remains difficult because it de-

pends on contrast, intent, and context. Annotation outcomes vary with elicitation method and annotator perspective (Oprea and Magdy, 2020; Casola et al., 2024), while model behavior is sensitive to domain and style (Jang and Frassinelli, 2024); multilingual settings introduce further variation (Casola et al., 2024) (§2.1).

LLM-generated text adds a source dimension to this task. Human-authored and AI-generated texts can each be sarcastic or non-sarcastic, so all four combinations are possible. A robust sarcasm detector should not infer sarcastic intent from source alone.

Machine-generated text detection (MGTD) imposes a contrasting representational demand: its objective distinguishes human-written from AI-generated text and therefore favors source-predictive representations (Guo et al., 2023; Wang et al., 2024) (§2.2). Some latent stylistic or distributional cues predictive of AI-generated text may overlap with, or become entangled with, features used for sarcasm detection. The output labels do not conflict; the potential conflict lies in how the two tasks organize and use the shared representation. Sarcasm detection should not rely on source differences alone, whereas MGTD must encode them. Shared-encoder multi-task learning can suffer task conflict and negative transfer (Mueller et al., 2022); more specifically, it can exploit knowledge useful for one task but spurious for another (Hu et al., 2022b). Evaluating the tasks only in separate models cannot reveal whether MGTD supervision changes the sarcasm decision.

We use joint training as a controlled intervention on the shared representation rather than as an assumed route to higher accuracy. A dual-head model lets us test how sarcasm decisions change when MGTD gradients also shape the shared representation. Architecturally matched single-head controls help isolate that change, and source $\times$ sar-

casm subgroups show where it appears.

As a downstream implication, we examine whether the same detectors provide reliable evaluation signals for generated sarcasm. ViSP, for example, derives PPO rewards from a sarcasm detector (Wang et al., 2026), while generated sarcasm may be judged for both sarcastic intent and human-likeness (§2.4). If detector scores are source-sensitive, they could favor source-predictive stylistic or distributional cues rather than measure sarcastic intent. We therefore compare detector scores for human authorship with source-blinded LLM ratings of perceived human-likeness separately in §6.

We investigate the interaction through matched comparisons. We train dual-head models on bilingual Chinese–English data and build DualTest, whose dual labels partition evaluation into four source $\times$ sarcasm quadrants. Compared with matched single-head controls, joint training substantially increases the sarcasm false-positive rate on AI-generated text prompted to be non-sarcastic. We refer to this auxiliary-task-induced degradation, concentrated in a particular source $\times$ sarcasm subgroup, as class-conditional negative transfer. The pattern persists with a second generator and is attenuated in both RCNN and interaction-projection intermediate variants.

We then examine the shared representation. Source information is highly decodable and correlates with sarcasm decisions. Representation interventions show that the subgroup gap is sensitive to a dominant-variance subspace, while leaving the causal content of that subspace unresolved (§5.3.2, Appendix E). Because probe accuracy alone does not establish how encoded information is used (Hewitt and Liang, 2019; Elazar et al., 2021), we treat the source-correlated shortcut as a mechanism hypothesis with preliminary support rather than an established cause.

Contributions. To our knowledge, we provide the first bilingual diagnostic study of how MGTD supervision changes sarcasm detection across Chinese and English, together with DualTest for source $\times$ sarcasm analysis. Matched comparisons across models and generators identify class-conditional negative transfer concentrated on AI-generated text prompted to be non-sarcastic; cross-architecture comparisons characterize how its severity varies across configurations. Representation analyses characterize and constrain

the source-shortcut hypothesis. Finally, an exploratory detector–judge comparison illustrates a downstream risk of using detector scores to evaluate generated sarcasm and perceived human-likeness.

2 Related Work

2.1 Sarcasm Detection and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016; Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. The task remains difficult in multilingual and class-imbalanced settings, especially because datasets differ in whether labels reflect intended sarcasm or annotator-perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024). Although scores are not directly comparable across datasets, languages, modalities, or evaluation setups, they illustrate substantial benchmark variability. On English iSarcasmEval (Abu Farha et al., 2022), the top two systems report positive-class F1 of 0.605 and 0.569; across MultiPICO languages (Casola et al., 2024), zero-shot ChatGPT positive-class F1 ranges from 0.152 to 0.567. On the bilingual multimodal SarcNet benchmark (Yue et al., 2024), a text-only XLM-R baseline reaches 0.734 F1.

Architecturally, this work follows Yang and Ikeda (2024), who compared mBERT, mBERT-BiLSTM, and mBERT-RCNN on mixed English–Arabic and English–Chinese training sets and found that the RCNN variant generally improved performance on two English test sets. Their RCNN implementation builds on the recurrent convolutional architecture of Lai et al. (2015); we therefore include RCNN among the intermediate architectures examined in §4.1.

The benchmark results summarized above come from different datasets, so §5.1 provides directly comparable mBERT dual-head baselines on DualTest. To our knowledge, existing sarcasm resources do not provide both sarcasm and human-versus-AI source labels. DualTest supplies both labels in a bilingual Chinese–English setting.

The Chinese and English source resources differ in domain, annotation procedure, and class balance, so bilinguality introduces additional distributional variation. We treat language as a robustness and stratification dimension rather than as the pri-

many transfer objective: the study asks whether the subgroup failure is confined to one language, not whether knowledge transfers across languages. We therefore report language-stratified results in addition to pooled metrics.

2.2 Machine-Generated Text Detection and Authorship Verification

Machine-generated text detection (MGTD) includes zero-shot model-likelihood methods, including probability-curvature detectors (Mitchell et al., 2023), and discriminative models trained on paired human–AI corpora (HC3; Guo et al., 2023). Generalization is a central difficulty: M4 (Wang et al., 2024) provides a benchmark across generators, domains, and languages and shows that detectors struggle on unseen domains or LLMs, while RAID (Dugan et al., 2024) stress-tests detectors under unseen generators, decoding variations, and adversarial attacks. This generalization gap is consistent with reliance on source-predictive regularities that are partly generator- or domain-specific; such cues are relevant here because they may enter a representation shared with sarcasm detection. Bevendorff et al. (2025) distinguish closed-set authorship attribution from open-set authorship verification and argue that realistic LLM detection is better cast as the latter. Our binary MGTD setup is narrower: we use it as a controlled source-classification auxiliary task, not as a measure of text quality.

In our literature search, we found no work that jointly trains sarcasm detection and MGTD with a shared encoder.¹ This gap motivates our study because source information has different roles in the two tasks: it is the target of MGTD but should not by itself determine a sarcasm detector’s prediction. Our study tests whether a shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, auxiliary tasks are not always beneficial: transfer depends on task relatedness and alignment between task data (Zhang et al., 2023; Wu et al., 2020), while gradient conflict is a central optimization challenge that can

contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Mueller et al. (2022) provide a relevant NLP precedent through matched single- and multi-task comparisons. They find similar negative-transfer patterns in canonical and text-to-text systems and note that models may infer task identity from input distributions, a strategy that may not transfer reliably out of distribution. He and Choi (2021) and Li et al. (2023) further analyze functional specialization in multi-task Transformer encoders. Shortcut learning offers a complementary reading: models may exploit features that are predictive in the training distribution yet fail under distribution shift (Geirhos et al., 2020; Ye et al., 2026). In MTL, shared representations can increase reliance on spurious correlations (Hu et al., 2022b; Chai et al., 2025). Related multi-dataset work shows that adding data can introduce spurious correlations and reduce worst-group performance (Compton et al., 2023).

Mitigation under a shared encoder takes two broad forms. Parameter-efficient methods introduce task-specific or task-conditioned lightweight components (AdapterFusion, Pfeiffer et al., 2021; MAD-X, Pfeiffer et al., 2020; HyperFormer++, Karimi Mahabadi et al., 2021; PALs, Stickland and Murray, 2019). Other methods explicitly gate or adapt information sharing within the shared computation (Wu et al., 2019; CA-MTL, Pilault et al., 2021).

Rather than isolating task-specific parameters, we examine whether alternative intermediate representations alter the subgroup failure induced by joint training. We compare a no-module baseline with two intermediate-module variants (§5.3). Unlike aggregate negative transfer, our analysis asks whether auxiliary-task supervision degrades performance only for particular source $\times$ sarcasm subgroups.

2.4 Misalignment in Detector-Based Evaluation and Rewards

The representational conflict studied above may also matter when detector outputs are reused beyond classification, particularly as evaluation or reward signals for generated text. Automated scorers, including LLM judges, are increasingly used to evaluate generated text (Zheng et al., 2023). Model-based scores are also optimized as rewards, but overoptimization can increase a proxy score while degrading a gold-standard measure (Gao et al., 2023). A detector can achieve useful clas-

¹Searches conducted in May 2026 and updated in July 2026 across Semantic Scholar, arXiv, ACL Anthology, and Google Scholar, using keyword combinations including “sarcasm detection,” “machine-generated text detection,” “AI-generated text detection,” “multi-task learning,” and “joint detection.”

sification accuracy without its continuous output being calibrated as a measure of style strength or generation quality. Multimodal sarcasm generation may face this risk: ViSP (Wang et al., 2026) uses the DIP sarcasm detector (Wen et al., 2023) as a PPO reward (Schulman et al., 2017). As an exploratory downstream analysis, Section 6 compares detector scores for human authorship with separately obtained, source-blinded LLM ratings of perceived human-likeness.

3 Dataset Construction

Testing for interference requires separate task supervision and evaluation labels for authorship and sarcasm. Here “source” denotes human-versus-AI authorship; “corpus” denotes an input dataset. We combine iSarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). None provides both labels. Mapping their annotations to a common binary sarcasm label yields 44,862 pooled instances (approximately 87% English), from which we construct three training sets and DualTest (Appendix A).

Sarcasm classification training set, SarcTrain (10,000, all human). Each language $\times$ label bucket contains 2,500 instances. Three buckets are downsampled; the Chinese sarcastic bucket starts from 2,396 records representing 2,394 unique texts and adds 104 oversampled records. The resulting set is organized into five folds.

MGTD training set, AuthTrain (19,995). From 9,999 language-balanced human seeds, DeepSeek-R1-0528-Qwen3-8B (Ollama tag `deepseek-r1:8b`) (DeepSeek-AI, 2025; Ollama, 2025) produces 9,996 retained outputs (99.97% pairing coverage); Appendix A details quality-control exclusions.

Four language $\times$ style templates follow the balanced seed labels. Authorship and requested-style labels are constructed as separate, approximately balanced factors; no independence is assumed for realized textual features or perceived sarcasm (Appendix A).

Generation training set, GenTrain (2,428; used in §6). From 5,257 eligible positive-sarcasm

records in iSarcasm and ToSarcasm, seed-42 sampling selects 2,103 records (1,214 English; 889 Chinese). Chinese oversampling adds 325 records, yielding 1,214 per language. The resulting set contains 374 rewriting, 839 response, and 1,215 comment instances and serves as the instruction-tuning data for SFT (Appendix A).

Diagnostic test set, DualTest (4,664). The human subset combines official iSarcasm (1,400 English) and ToSarcasm (972 Chinese) test sets. Generation quality control retains AI counterparts for 2,292 of 2,372 texts (96.6% coverage). Human texts retain their annotations; for AI texts, the sarcasm label records the prompted condition rather than an independently annotated judgment of perceived sarcasm.

The four source $\times$ sarcasm groups contain 823 human-sarcastic, 1,549 human-non-sarcastic, 779 AI-sarcastic, and 1,513 AI-non-sarcastic instances. Appendix A gives the language $\times$ source $\times$ sarcasm distribution.

4 Dual-Head Detection Model

4.1 Tasks and Architecture

Given a Chinese or English input text x , the model produces two separate, non-mutually-exclusive binary predictions: whether x is labeled sarcastic and whether its authorship source is labeled human or AI. Any combination of the two labels is valid. The tasks share a single DeBERTa-v3-large encoder (He et al., 2023), allowing us to test whether MGTD training changes the representation read by the sarcasm head. This encoder was pretrained on English corpora; §5.1 directly compares it with multilingual-pretrained mBERT / XLM-R on the same bilingual test set. Encoder outputs are aggregated by an intermediate module and then passed to two task-specific classification heads; Figure 1 shows the RCNN configuration.

The RCNN configuration uses BiLSTM + CNN as its intermediate module. A single-layer bidirectional LSTM with hidden size 256 per direction produces a 512-dimensional output at each token position. We concatenate it with the 1,024-dimensional encoder output, then apply three groups of one-dimensional convolutions with kernel widths 3/4/5 and 128 channels each. Global max pooling produces a 384-dimensional vector z . After dropout, z is passed to the two heads. This module adapts the mBERT-RCNN intermediate ar-

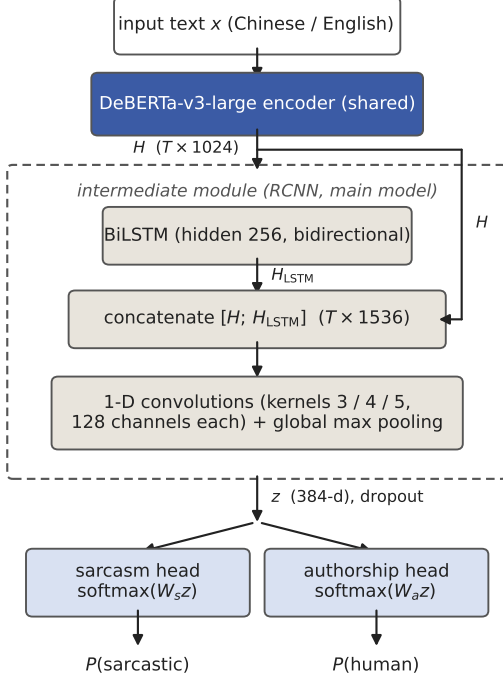


Figure 1: RCNN dual-head configuration: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with task-branch interaction projection, while retaining the same encoder backbone and task definitions (Appendix B).

chitecture of Yang and Ikeda (2024) to a DeBERTa-based dual-head setting.

To test the intermediate layer’s role, §5 compares two variants. The plain model sends [CLS] directly to both heads. The interaction-projection model forms two learned 256-dimensional projections, h_s and h_a , and updates them as $h'_s = h_s + P_{a \rightarrow s}(h_a)$ and $h'_a = h_a + P_{s \rightarrow a}(h_s)$. Each branch contains one vector, so no query-key interaction remains; P is the effective learned value/output projection. The configurations retain the same encoder and task definitions while testing distinct intermediate-representation designs (Appendix B).

4.2 Training Protocol

Training uses a mixed loader that supplies task-homogeneous batches according to a 2:1 MGTD:sarcasm schedule, matching the approximate data-size ratio between AuthTrain and SarcTrain. Let the shared parameters be θ_{sh} (the encoder and intermediate module), let $z(x)$ be their

aggregated representation for input x , and let W_s and W_a denote the two classification heads. At step t , the loss is computed only for the head corresponding to the task of the current batch B_t :

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}_w(W_s z(x), y), & B_t \subset \text{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}(W_a z(x), y), & B_t \subset \text{AuthTrain}, \end{cases} \quad (1)$$

where CE_w is a pre-specified cost-sensitive cross-entropy for sarcasm (weights are provided in Appendix B); because SarcTrain is class-balanced, the weighting emphasizes errors on sarcastic examples rather than correcting training-set imbalance. Softmax is applied to logits only when probabilities are required. The gradient flow follows from the equation: on sarcasm batches, $\partial \mathcal{L} / \partial W_a = 0$ (symmetrically, $\partial \mathcal{L} / \partial W_s = 0$ on MGTD batches), whereas θ_{sh} receives gradients from both types of batches. Unless otherwise stated, binary predictions use a fixed probability threshold of 0.5 rather than a threshold tuned on DualTest.

The encoder uses a lower learning rate than newly added modules. The sarcasm-head rate is fixed across configurations; MGTD-head rates are reported in Appendix B.

Model selection accounts for both tasks. At epoch e , the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}_{\text{stop}}^{(e)} = \frac{1}{2} \frac{\mathcal{L}_s^{(e)}}{\mathcal{L}_s^{(1)}} + \frac{1}{2} \frac{\mathcal{L}_a^{(e)}}{\mathcal{L}_a^{(1)}}, \quad (2)$$

where $\mathcal{L}_s^{(e)}$ and $\mathcal{L}_a^{(e)}$ are the epoch- e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics.

Five runs are defined by the SarcTrain validation folds: four folds train and one validates sarcasm, while AuthTrain uses a fixed 90/10 split. Each run reloads the pretrained encoder and randomly initializes new modules and heads. Configurations share fold assignments and nominal seeds, enabling split-matched comparisons but not identical initialization.

DualTest enters no training or validation split. Five-run standard deviations combine fold and initialization variation. Wilson intervals are conditional on a fixed final model and cover test-

instance, not training or generator, uncertainty (Table 3; Appendix C.7).

Each subgroup configuration has a final model trained on all applicable data with a fixed epoch count determined from the five validation runs, an independent seed, and no early stopping. Their weights and training duration are not selected using DualTest, and they are not used in five-run aggregates; §6 uses the final RCNN dual-head model (Appendix B).

5 Experiments and Diagnosis

The experiments move from aggregate comparison to matched ablation and subgroup diagnosis. We first compare pretrained encoders and intermediate configurations, then measure the change produced by adding MGTD training, and finally partition predictions into source $\times$ sarcasm quadrants. DualTest is used for these descriptive and diagnostic comparisons, but not for training, validation, threshold selection, or epoch selection. Aggregate results are five-run means $\pm$ standard deviations; fixed final models support paired instance-level subgroup comparisons and Wilson intervals.

5.1 Backbone Comparison

We compare dual-head mBERT, XLM-RoBERTa-base, DeBERTa-v3-base, and DeBERTa-v3-large under one evaluation protocol, adding the two intermediate variants on the large encoder. Their pretraining spans multilingual Wikipedia (Devlin et al., 2019), multilingual CommonCrawl (Conneau et al., 2020), and primarily English DeBERTa-v3 pretraining with an ELECTRA-style objective (He et al., 2023; Clark et al., 2020). Positive-class F1 uses the sarcastic class for sarcasm and the human class for MGTD on full DualTest.

Table 1 shows higher mean F1 for the evaluated DeBERTa configurations than for mBERT and XLM-R, with RCNN obtaining the highest means. The intermediate variants also show 34–47% lower observed standard deviations across the five runs. Because backbone scale, pretraining, and intermediate representation vary together, this comparison characterizes the evaluated configurations rather than isolating one factor. RCNN reaches 0.6309 sarcasm F1 and 0.9301 MGTD F1, establishing a strong shared-encoder diagnostic configuration while leaving a substantial task-level performance gap to explain. Its sarcasm F1 is 0.819 in Chinese

Model	Sarc. F1	MGTD F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLM-RoBERTa-base dual-head	0.4603 $\pm$.057	0.8321 $\pm$.025
DeBERTa-v3-base dual-head	0.5646 $\pm$.056	0.8952 $\pm$.004
DeBERTa-v3-large plain dual-head	0.6126 $\pm$.031	0.9221 $\pm$.010
+ interaction projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN	0.6309 $\pm$.020	0.9301 $\pm$.005

Table 1: Descriptive dual-head results across backbones and architectural variants (DualTest; five-run mean $\pm$ standard deviation; bold = highest mean).

Arch.	Task	Dual	Single	Δ
Plain	Sarc.	0.6126 $\pm$.031	0.6734 $\pm$.010	−6.08
Plain	MGTD	0.9221 $\pm$.010	0.9054 $\pm$.027	+1.67
RCNN	Sarc.	0.6309 $\pm$.020	0.6751 $\pm$.022	−4.43
RCNN	MGTD	0.9301 $\pm$.005	0.9253 $\pm$.006	+0.49

Table 2: Dual-head vs. single-head models (mean $\pm$ standard deviation over five fold-based runs; Δ = dual – single, in percentage points).

and 0.429 in English; prevalence, genre, annotation, and language jointly shape this contrast. Appendix C provides macro-averaged, human-subset, and language-stratified results.

5.2 Dual-Head versus Single-Head Ablation

Within each architecture, dual- and single-head models share the encoder, intermediate structure, sarcasm data, fold assignments, and sarcasm-specific hyperparameters; dual runs additionally receive AuthTrain batches and MGTD updates. Single-head controls traverse the full sarcasm fold each epoch, whereas dual-head models draw sarcasm batches under the 2:1 schedule (§4.2).

Joint training lowers mean sarcasm F1 by 6.08 pp for plain and 4.43 pp for RCNN, whereas the corresponding MGTD changes are +1.67 and +0.49 pp. All ten paired sarcasm differences are negative (−0.8 to −8.6 pp), so the transfer cost recurs across every split-matched run and both architectures. This repeated asymmetry identifies sarcasm as the task bearing the aggregate cost and motivates locating that cost by source and class. The smaller dual–single gap under RCNN further motivates the intermediate-architecture diagnosis below; capacity and compute differences are recorded in Limitations.

5.3 Four-Quadrant Diagnosis: Class-Conditional Negative Transfer across Intermediate Architectures

Aggregate F1 hides whether errors depend jointly on source and sarcasm class. We therefore partition DualTest by source $\times$ sarcasm. Human labels are dataset annotations and AI labels record prompted conditions; single-class quadrants use recall for sarcastic subsets and false-positive rate for non-sarcastic subsets.

5.3.1 Three Architectures and Single-Head Controls

The three dual-head variants show how the subgroup pattern changes with intermediate representation. Plain and RCNN single-head models provide matched controls for adding AuthTrain and MGTD supervision. Because these controls retain the same all-human sarcasm data, they distinguish the effect of joint training from the absence of AI texts in SarcTrain itself. Table 3 reports sarcasm predictions; corresponding MGTD results are in Appendix C.

The plain dual-head model predicts sarcasm for 86.3% of AI $\times$ non-sarcastic instances. Interaction projection and RCNN reduce this rate to 40.9% and 34.1%, respectively, while both matched single-head controls yield 19.3%. Thus, the two intermediate configurations reduce the observed severity by more than 45 pp relative to the plain dual-head model.

On the same texts, plain dual versus single gives 1,021/7 discordant pairs ($p < .001$). The matched single-head model receives the same all-human sarcasm supervision but not MGTD training, locating the large increase in the joint-training procedure. RCNN’s matched comparison shows the same direction with a smaller gap; interaction projection has no single-head counterpart and contributes complementary configuration-level evidence.

No analogous extreme subgroup error appears in the MGTD head: its false-positive rate is 6.1% for plain and 6.6% for RCNN. Within AI texts, sarcasm AUC remains 0.82–0.88 while joint training raises score levels and largely preserves ordering. The failure is therefore not a collapse of within-AI ranking, but a source-conditioned upward shift that produces class-conditional negative transfer on the non-sarcastic prompted subset (Appendix C.7).

The matched plain comparison makes the concentration especially clear: its non-sarcastic false-

positive rate changes by 4.2 pp on human text but by 67.0 pp on AI-generated text. The aggregate F1 decrease is therefore driven by a sharply localized source $\times$ class effect rather than a uniform loss across DualTest.

5.3.2 Mechanism Hypothesis and Representation-Level Diagnostics

We hypothesize that MGTD-shaped representations expose source-correlated cues to a sarcasm head that receives no sarcasm supervision on AI text. Source is highly decodable in both dual-head (AUC 0.986) and matched single-head (0.974) representations, so its presence alone is not the explanation. In the dual-head model, however, the MGTD-head AI logit correlates with the sarcasm-head logit on the target subgroup ($\rho = 0.519$), while both MGTD-head and probe directions align with the sarcasm direction in whitened geometry.

Representation interventions further characterize the effective variation. LEACE (Belrose et al., 2023) reduces measured linear source decodability from AUC 0.982 to 0.589 and shifts the non-sarcastic false-positive rates from 84.8%/32.8% on AI/human text to 68.7%/49.5%. Removing a high-variance direction orthogonal to the estimated linear source direction yields 66.7%/54.1% while preserving measured source decodability; ten sampled random controls show no comparable source-specific shift. Under LEACE, the unweighted mean across the two source groups remains nearly unchanged (58.8% to 59.1%), showing error redistribution rather than overall mitigation. Together, these results support a bounded source-correlated shortcut hypothesis and show prediction sensitivity to high-variance directions without isolating a unique causal feature (Appendix E).

This evidence chain goes beyond probe accuracy alone: task-head logit correlation connects source-related behavior to the decision, targeted interventions change how errors are allocated across sources, and the orthogonal and random controls constrain the interpretation. The result is a representation-level diagnosis of score sensitivity rather than a causal localization inferred from output labels alone.

5.3.3 Cross-Generator Replication

We regenerate only DualTest’s AI side with Qwen3.5-9B (Ollama tag qwen3.5:9b) (Qwen Team, 2026), retaining the human texts and trained detectors. Among 1,550 non-sarcastic-condition

Model	H sarc. rec.	H non-sarc. FP	AI sarc. rec.	AI non-sarc. FP
Plain dual-head	67.1%	32.9%	98.2%	86.3% [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	40.9% [38.5, 43.4]
RCNN dual-head	80.8%	40.5%	89.3%	34.1% [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	19.3% [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	19.3% [17.4, 21.4]

Table 3: Five sarcasm classifiers on four source $\times$ sarcasm quadrants (recall / false-positive rate, FP). Human labels are annotations; AI labels record prompted conditions. Brackets give model-conditional Wilson 95% intervals; full intervals and paired tests are in Appendix C.7.

outputs, plain dual rises from 86.3% to 91.3% and its matched single falls from 19.3% to 7.9%, increasing the gap from 67.0 to 83.4 pp. RCNN and interaction projection remain lower at 22.1% and 29.5%. Thus, both generators preserve the large plain dual–single separation and the lower rates of the two intermediate configurations. The second generator reproduces the gap direction with a different magnitude (Appendix C.7).

6 Exploratory Comparison of Detector Scores and LLM-Judge Ratings

As a downstream implication, we test whether source-classification probability tracks perceived human-likeness and whether sarcasm probability tracks LLM ratings. Human-authorship probability and perceived human-likeness are distinct constructs; their comparison directly tests whether a classifier score can function as a generation-evaluation proxy. We compare 600 outputs from Qwen3-8B (Yang et al., 2025) before and after QLoRA fine-tuning on GenTrain. The RCNN detector and three LLM judges (DeepSeek-AI, 2024; Comanici et al., 2025; Singh et al., 2026) score outputs from 180 Chinese topic-comment and 120 English rewriting prompts; the judges are not shown the ground-truth source, and language is coupled with generation task.

Agreement varies by dimension and language. In Chinese, SFT raises detector sarcasm from 0.780 to 0.895 but lowers mean judge rating from 4.01 to 3.71; sarcasm correlation is -0.020 in Chinese and $+0.338$ in English. SFT outputs also receive higher human-authorship probabilities but lower perceived human-likeness ratings, yielding a pooled correlation of -0.212 . Within-condition correlations show that this negative association is driven mainly by the reversal between base and SFT conditions.

This condition-level reversal is directly relevant to generation evaluation: fine-tuning moves the de-

tector scores in the apparently favorable direction while moving the external LLM ratings downward. A detector can therefore provide a useful classification boundary without its continuous score preserving the ordering required of a style or quality measure.

Inter-judge agreement is low to moderate at the instance level, but five of six language $\times$ judge comparisons assign lower sarcasm ratings to SFT than to the base outputs. A plain-detector control changes the reported correlations by at most 0.02. The consistent condition-level directions therefore show that detector probabilities require external validation as generation-quality proxies or rewards. Full protocols, tests, and representative cases are reported in Appendices C–D.

7 Conclusion

Source and sarcasm labels can coexist, yet joint training can change the sarcasm decision. Relative to a same-architecture single-head control using the same sarcasm data, the plain dual-head model has a sharply higher false-positive rate on AI text generated under non-sarcastic instructions. A second generator reproduces the gap’s direction but not its magnitude, and both intermediate configurations have lower rates than the plain dual-head model. Representation analyses link the shift to source-correlated, high-variance directions, while revealing error redistribution rather than overall improvement and leaving causality unresolved. Detector probabilities and LLM-judge ratings also differ by dimension and task–language setting, so the evaluated scores require external validation as measures of style realization or quality. In our setting, aggregate metrics should be supplemented with authorship-source $\times$ reference-class analysis.

Limitations

DualTest combines original sarcasm annotations for human texts with instruction-conditioned style labels for AI texts. Because instruction compliance is not independently annotated, an apparent AI-side false positive can reflect detector error or realized sarcasm despite a non-sarcastic instruction. Generation failures also leave 80 human texts without retained main-generator counterparts, so source-side comparisons are not fully paired. Seven exact train–test matches remain on the AI side (two from SarcTrain and five from AuthTrain); their influence cannot be separated without retraining.

The MGTD training text is produced with one generator, and the second generator replaces only DualTest’s AI side. The results therefore establish a directional replication across two retained output sets rather than coverage of broader model families, scales, or training paradigms. The input corpora also operationalize sarcasm differently, and the detector does not use conversational context.

DualTest supports both the descriptive architecture comparison and the subsequent subgroup diagnosis, so cross-configuration results are descriptive rather than an independent model-selection test. The sarcasm loss also upweights the positive class despite balanced SarcTrain; this cost-sensitive choice can affect absolute false-positive rates, although each matched dual–single comparison uses the same weight. The dual and single procedures share architecture, sarcasm data, folds, and task-specific learning rates, but differ in task schedule, total shared-encoder updates, and early stopping; they estimate the complete joint-training procedure rather than a compute-matched auxiliary-gradient effect. Cross-architecture comparisons are neither capacity- nor optimization-matched, their MGTD-head learning rates differ, and interaction projection lacks a single-head counterpart.

The mechanism evidence is primarily behavioral and linear and uses one final plain dual-head model. Source is also highly decodable from the single-head representation, so decodability alone does not explain the joint-training-specific shift. The orthogonal high-variance control prevents attribution of the intervention effect specifically to source identity, and the interventions mainly redistribute non-sarcastic errors from AI to human text rather than lowering their source-averaged rate.

The generation analysis compares human-authorship probability with perceived human-likeness, which are related but non-equivalent constructs. It uses three LLM judges rather than human evaluators, and their instance-level agreement is low to moderate. The Chinese and English subsets differ in sample size, generation task, prompt structure, and source data. The results therefore show condition-level directional differences between the evaluated detector scores and selected LLM-judge ratings, not human-perceived quality or a pure language effect.

Priority extensions are independent annotation of AI-side style realization, capacity- and training-budget-matched controls, and multi-generator, multilingual, and calibration analyses. Reward-based generation and nonlinear causal tests provide subsequent directions.

Ethical considerations

All human-authored texts come from publicly released research datasets: iSarcasm, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. Explicit repository licenses were available for iSarcasm and CSC; for sources without an identified redistribution license, we do not redistribute the human-authored text (Appendix A). We collect no new user data. We did not conduct an additional personally identifiable information or offensive-content audit beyond the preparation reflected in the released corpora.

All AI texts were generated locally using publicly released open-weight models (DeepSeek-R1-0528-Qwen3-8B and Qwen3.5-9B). They are used only for training and evaluation and are labeled as AI-generated. The study recruits no new human participants, but analyzes human-authored texts from previously released datasets. The automated evaluation transmits each evaluation input, generated output, and rating instructions to the DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano APIs; it does not transmit dataset metadata or ground-truth source labels.

The detectors are developed for research diagnosis. Their outputs should not be used as evidence of authorship or misconduct at the individual-text level without human review and domain-specific validation, nor deployed for content moderation or automated sanctions without evaluating out-of-distribution behavior. The class-conditional failure diagnosed here is an example of systematic sub-

group error affecting a particular type of generated content.

AI-assisted tools were used for grammar correction, wording suggestions, and formatting. They did not generate the reported experimental outputs or statistics; the authors determined the study design, analysis, and conclusions and take full responsibility for the work.

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A Detailed Dataset Statistics and Quality Checks

Corpus	Lang.	Inst.	Use
iSarcasm (SemEval-2022 T6)	English	3,467	Sarc. train + SFT
CSC	English	6,878	Sarc. train
ToSarcasm	zh 3,892 / en 6	3,898	Sarc. train + SFT
News Headlines v2	English	28,619	Sarc. train
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	Sarc. train

Table 4: Input corpora. Genre / annotation: iSarcasm — tweets, speaker self-annotation; CSC — conversational responses; ToSarcasm — topic comments, third-party annotation; News Headlines — distant supervision (The Onion / HuffPost); Open Chinese Corpus — Weibo, third-party annotation.

A.1 Corpus details. The pooled corpus collection contains 44,862 instances in total (86.9% English / 13.1% Chinese). Chinese–English 1:1 balance in SarcTrain is achieved through bucketed sampling. The two English buckets and the Chinese non-sarcastic bucket contain more than 2,500 available instances and are randomly downsampled within each bucket. The Chinese sarcastic bucket contains 2,396 source records representing 2,394 unique texts; 104 records are added by oversampling to reach 2,500. Including source-level duplicate text, the final bucket’s text-duplication rate is 4.24% (A.3).

Licensing and release scope. The iSarcasm and CSC repositories state MIT licenses. We did not identify explicit redistribution licenses in the releases we accessed for ToSarcasm, News Headlines v2, or the Open Chinese Internet Sarcasm Corpus. We therefore do not redistribute human-authored text from those sources. Planned research artifacts are limited to eligible AI-generated text, labels and metadata, and reconstruction scripts that require users to obtain the human-authored corpora from their original sources.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language $\times$ source $\times$ sarcasm distribution of DualTest. Human labels are dataset annotations; AI labels record prompted conditions.

A.2 Three-dimensional distribution of DualTest.

A.3 Data quality checks. Automated checks confirm language and label balance across the five SarcTrain folds. AuthTrain contains 9,996 complete human–AI pairs and three unmatched human instances. Exact-match screening finds two SarcTrain instances (0.02%) and five AuthTrain instances (0.025%) that also occur on the AI side of DualTest; overlap on the human subset is zero. These seven residual matches are disclosed because they occur on the AI side used in the central subgroup analysis. SarcTrain’s text-duplication rate is 0.34% for English / 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic bucket (4.24%; non-sarcastic bucket: 0.08%) and results from the oversampling used to fill that bucket (A.1). Post-processing also checks for the explicit chain-of-thought markers and reasoning prefaces described in A.4 (Wei et al., 2022).

A.4 AI generation protocol. All AI texts are produced by locally deployed generation models through the Ollama API (DeepSeek-R1-0528-Qwen3-8B under tag deepseek-r1:8b for the main results; Qwen3.5-9B under tag qwen3.5:9b for cross-generator replication), using temperature = 0.8, top- p = 0.95 (Holtzman et al., 2020), and a generation limit of 500 tokens. Generation instructions use four templates defined by language $\times$ style, with style determined by the seed instance’s

sarcasm label. The templates constrain output style and length (20–80 Chinese characters; 10–50 English words). This constraint is intended to reduce, but does not eliminate, source differences in length; [Appendix E](#) separately repeats the representation correlation analysis within length tertiles. The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

【任务】生成一条讽刺风格的评论
 【主题】{seed}
 【要求】
 - 直接给出简短的评论，不要解释思路或分析
 - 风格要讽刺、幽默、夸张
 - 表达与主题相关的观点即可，不必追求严谨
 - 20–80 字为佳
 【评论】

Its pinyin transliteration is:

[rènwù] shēngchéng yī tiáo fěngcì
 fēnggé de pínglùn
 [zhǔtí] {seed}
 [yāoqiú]
 - zhíjiē gěichū jiǎnduǎn de pínglùn,
 bù yào jiěshì sīlù huò fēnxī
 - fēnggé yào fěngcì, yōumò, kuāzhāng
 - biǎodá yǔ zhǔtí xiāngguān de guāndiǎn
 jíkě, bùbì zhuīqiú yánjīn
 - 20–80 zì wéi jiā
 [pínglùn]

Its English translation is:

[Task] Generate a sarcastic-style
 comment
 [Topic] {seed}
 [Requirements]
 - Directly provide a short comment;
 do not explain your reasoning or
 analysis
 - The style should be sarcastic,
 humorous, and exaggerated
 - It is sufficient to express a
 viewpoint related to the topic;
 strict rigor is not required
 - Preferably 20–80 Chinese
 characters
 [Comment]

The Chinese non-sarcastic template changes only the style line to “The style should be normal and natural.” The English templates follow the same structure, using either “sarcastic, humorous, exaggerated” or “normal, natural,” with a length limit of 10–50 words.

Post-processing removes `<think>...</think>` segments, reasoning prefaces such as “好的” (*hǎo de* ‘okay’), “嗯” (*ní* ‘hmm’), “Okay,” “Let me,” and

“First,” and Markdown markup. Outputs that remain overlength or cyclic after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The Qwen3.5-9B replication has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. Generator-specific rates therefore use each retained full output set rather than an identical seed intersection. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from DeepSeek-R1-0528-Qwen3-8B.

A.5 SFT training and generation-evaluation set.

Eligibility filtering retains 5,257 positive-sarcasm records suitable for generation tasks (2,972 English; 2,285 Chinese). A seed-42 shuffle and 40% sampling select 2,103 records (1,214 English; 889 Chinese); language balancing then adds 325 oversampled Chinese records, producing GenTrain (2,428; 1,214 per language). By prompt type, it comprises 374 rewriting, 839 response, and 1,215 comment instances. Generation evaluation uses an independent set of 300 prompts (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. The detector scoring and three source-blinded LLM evaluations in §6 are both conducted on this set.

B Hyperparameters and Training Details

Group	Enc.	Inter.	Sarc.	MGTD
Plain / backbone	5e-7	—	5e-6	1e-5
Intermediate variants	5e-7	5e-6	5e-6	5e-6
Single-head	5e-7	5e-6 [†]	5e-6	1e-5

Table 6: Discriminative learning rates. [†]RCNN single-head only.

B.1 Discriminative learning rates. The sarcasm-head learning rate is fixed at 5e-6 across configurations. The MGTD-head rate is 1e-5 for the plain dual-head and single-head models and 5e-6 for the intermediate variants. Cross-configuration differences may therefore reflect optimization settings as well as intermediate-representation design. The matched dual-single

comparisons retain the same settings within each architecture.

B.2 Shared training hyperparameters. Training uses AdamW with default PyTorch betas (0.9, 0.999) and $\epsilon = 10^{-8}$, weight decay 0.01, and parameter-group learning rates in B.1. A LambdaLR schedule linearly warms up for 200 steps and then keeps the learning rate constant. Training uses at most 10 epochs, early-stopping patience 5, and gradient clipping 1.0. The physical batch size is 4 with gradient accumulation 4, giving an effective batch size of 16. The MGTD:sarcasm sampling ratio is 2:1, and sarcasm class weights are [1.0, 2.5]. Both choices were specified before testing. Because SarcTrain is balanced, the class weights implement cost-sensitive emphasis rather than imbalance correction. The normalized joint validation loss is $0.5 \times [\text{sarcasm validation loss} / \text{first-epoch value}] + 0.5 \times [\text{MGTD validation loss} / \text{first-epoch value}]$. The single-head controls traverse the full sarcasm training fold each epoch, whereas the dual-head models draw sarcasm batches under the 2:1 task-mixing schedule.

All fold-based runs start from seed 42 plus the fold index and share the same fold assignments and nominal seed scheme across architectures and ablations. MGTD data use a fixed 90/10 holdout in every fold. For each final-model configuration, the fixed epoch count is the mode of its five best epoch counts plus one, capped at 10 epochs. The final model uses all applicable training data, seed +1000, no validation set, and no early stopping. Main tables report five-run means and standard deviations rather than final-model point estimates.

Model sizes, software, and hardware environment. The encoders have approximately 178M (mBERT), 278M (XLM-RoBERTa-base), 184M (DeBERTa-v3-base), and 435M (DeBERTa-v3-large) parameters; the SFT generator Qwen3-8B has about 8B parameters, of which QLoRA updates only low-rank adapters. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB); the five-fold training of the backbone comparison, architecture, and ablation variants together with the final model amounts to on the order of tens of GPU-hours, with a single detector run taking from a few minutes to a few hours and SFT about two hours. The software environment is PyTorch 2.9.0 (CUDA 12.8), transformers 4.57.1, and peft 0.18.0. AI texts are generated locally through Ollama (Appendix A.4);

blinded LLM evaluation uses API calls made in May 2026 (Appendix C.3).

B.3 SFT (QLoRA). We fine-tune Qwen3-8B using QLoRA (Detmiers et al., 2023), with 4-bit loading and bfloat16 computation. The LoRA adapters (Hu et al., 2022a) use rank 64, alpha 16, dropout 0, and target q/k/v/o_proj plus gate/up/down_proj. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses learning rate $5e-5$, three epochs, batch size 1, gradient accumulation 8, and warmup ratio 0.05. After language balancing, a seed-42 split holds out 15% of GenTrain for validation; validation loss selects the checkpoint.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). RCNN macro-averaged F1, under the same five-run protocol and calculated from each run’s confusion matrix, is 0.6678 ± 0.036 for sarcasm and 0.9298 ± 0.005 for MGTD. Sarcasm macro-F1 is higher than its positive-class F1 (0.6309) because the majority class has a higher F1. These values supplement rather than rerank the full positive-class comparison in §5.1.

C.2 Complete four-quadrant results for the MGTD head (including single-head controls). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H×s	H×n	AI×s	AI×n
Plain dual	91.4	91.5	3.6	6.1
Inter.-proj. dual	85.4	85.4	1.8	4.1
RCNN dual	94.8	93.2	5.5	6.6
Plain single	90.4	86.6	4.1	4.2
RCNN single	90.5	92.4	3.6	5.7

Table 7: MGTD-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

The single-head MGTD controls remain within single-digit percentage points of the dual-head models in every quadrant. No analogous extreme class-conditional error appears on the MGTD side, consistent with §5.3.

No comparable subgroup increase appears on the qwen3.5:9b replication set: AI × non-sarcastic false-positive rates are 0.9%, 1.3%, and 0.5% for plain, interaction projection, and RCNN; AI × sar-

castic rates are at most 0.4%; and overall MGTD F1 scores are 0.952, 0.917, and 0.966.

C.3 Blinded-evaluation protocol, inter-judge agreement, and group-level tests (§6). Each sample is sent separately to the three judge APIs with one prompt (May 2026; temperature = 0.3; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated text, not the ground-truth source. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. Base and SFT outputs use the same prompt and are evaluated separately.

Pairwise instance-level agreement is moderate to low. Spearman ρ ranges from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). The conclusion in §6 therefore rests on consistent condition-level directions, not agreement in instance rankings.

Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) show that SFT receives lower sarcasm ratings than the base model in five of six language $\times$ judge groups. The pooled correlations in Table 8 use the three-judge mean as a descriptive aggregate; because inter-judge agreement is limited, the judge-specific results remain necessary.

Subset	n	Sarc. ρ	MGTD ρ
Overall	600	+0.136*	−0.212*
Chinese	360	−0.020	−0.218*
English	240	+0.338*	−0.270*

Table 8: Spearman correlations between detector probabilities and LLM-judge ratings (RCNN detector, $n = 600$ pooled over base and SFT). * significant; other entries not significant.

C.4 Paired tests between SFT and human references (supplementary). This test is performed on the Chinese subset of the generation-evaluation set ($n = 180$; GPT-5-nano is missing three instances, leaving 177). Directions are inconsistent: for DeepSeek-V3, SFT is significantly below the reference on human-likeness (-0.161 , $p = 0.0025$); for GPT-5-nano, SFT is significantly above the reference on sarcasm ($+0.186$, $p = 0.0114$); the other four groups are not significant.

C.5 Plain-detector control. With the plain detector, Chinese sarcasm changes from 0.765 to

Judge / lang.	n	SFT	Base	adj. p
DeepSeek-V3 / en	120	3.292	4.025	<.0001
DeepSeek-V3 / zh	180	3.878	4.044	.0008
Gemini-2.5-FL / en	120	3.750	3.992	.0008
Gemini-2.5-FL / zh	180	3.994	4.028	.2008*
GPT-5-nano / en	119	3.681	4.084	<.0001
GPT-5-nano / zh	180	3.517	3.911	<.0001

Table 9: Group-level tests for blinded sarcasm judgments (Wilcoxon + BH-FDR). * not significant. GPT-5-nano lacks one English base rating (C.3: $n = 119$) and three Chinese reference ratings (affecting C.4, not the C.3 Chinese row); tests use pairwise deletion.

0.763 (-0.001), English sarcasm from 0.843 to 0.659 (-0.184), and the MGTD score rises by $+0.760$ in Chinese and $+0.705$ in English. The two detectors therefore agree on the rise in MGTD scores, while the sarcasm direction depends on the detector.

Correlations are $+0.142$ for sarcasm and -0.214 for MGTD overall; -0.037 (not significant) and -0.209 in Chinese; and $+0.342$ and -0.283 in English. Each differs from the RCNN-detector value by at most 0.02. Negative human-authorship-probability versus human-likeness associations are similar across the two detectors, whereas the condition-level sarcasm-score change is architecture-dependent, especially in Chinese.

Within-condition correlations (RCNN detector). For MGTD, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese $+0.10$, English -0.09). The SFT condition has $\rho = -0.12$ ($p = 0.039$; Chinese -0.19 , $p = 0.009$; English -0.12 , not significant). The pooled value of -0.212 is driven mainly by the opposing condition-level changes: SFT raises the detector score by about 0.67 while lowering the mean blinded rating. There is no positive detector–judge alignment on this dimension within conditions.

For sarcasm, within-condition correlations are $+0.05$ for Chinese base (not significant), $+0.15$ for Chinese SFT ($p = 0.048$), $+0.41$ for English base, and $+0.37$ for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.6 Detector scores before and after SFT across the two task–language settings. Scores from the RCNN detector for base and SFT outputs (0–1):

Lang.	Cond.	Sarc.	MGTD
Chinese	Base → SFT	0.780 → 0.895	0.289 → 0.969
English	Base → SFT	0.854 → 0.730	0.198 → 0.864

Table 10: Detector scores before/after SFT (RCNN detector).

Sarcasm scores move in opposite directions in the Chinese comment-generation and English rewriting settings: +0.115 and −0.124, respectively. GenTrain itself is balanced at 1,214 instances per language; the evaluation settings instead differ in task, prompt format, source data, and sample size, all of which may contribute to the contrast. MGTD scores rise sharply in both settings, so the detector assigns SFT outputs higher probabilities of human authorship.

Mean LLM judgments for sarcasm / human-likeness are 3.77/4.17 for human references, 4.01/4.24 for the base model, and 3.71/4.08 for SFT. These directions conflict with the detector’s MGTD score, as analyzed in §6.

C.7 Conditional test-instance uncertainty: confidence intervals, paired tests, and subset F1. Conditional on each fixed final model, we quantify test-instance uncertainty with Wilson 95% intervals for quadrant rates and 95% percentile intervals from 10,000 instance-level bootstrap samples for F1. These intervals do not include uncertainty from training, generator sampling, corpus construction, or human–AI pairing. They complement the between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3.

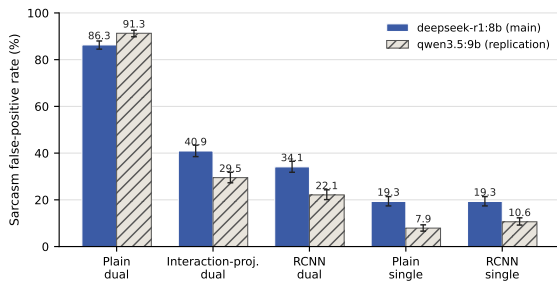


Figure 2: Sarcasm false-positive rates on the AI × non-sarcastic subset for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

The fixed final-model configurations show the same directional dual–single gap with the second

generator, although its magnitude differs.

Model	deepseek	qwen
Plain dual	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Inter.-proj. dual	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on the AI × non-sarcastic subset (two generators; %).

Paired McNemar tests use predictions on the same AI × non-sarcastic texts from the main-generator set. Discordant pairs are 1,021 / 7 for plain dual vs. plain single, 239 / 15 for RCNN dual vs. RCNN single, 805 / 15 for plain dual vs. RCNN dual, and 203 / 100 for interaction projection vs. RCNN dual (all $p < .001$). The discordant counts and absolute rate differences carry the substantive effect size.

Main-model MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the source label is constant within the human subset, positive-class F1 reduces to recall and is not reported separately.

Chinese sarcasm F1 is higher than English sarcasm F1 for the RCNN model (0.819 vs. 0.429). The contrast is confounded by prevalence, genre, annotation procedure, and language: sarcasm prevalence is 63.8% in Chinese but 14.2% in English, and the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval whereas the Chinese side uses topic-comment-style ToSarcasm. The positive-class F1 values are therefore not directly comparable as a pure language contrast.

DeBERTa-v3’s SentencePiece vocabulary (Kudo and Richardson, 2018) tokenizes Chinese largely by character, at approximately 1.02 tokens per character with 2.5% out-of-vocabulary symbols. This observation does not separate language from the other dataset differences above.

The separately retrained final RCNN model’s overall sarcasm F1 is 0.663, compared with 0.629 on human texts alone, because the overall score also includes AI texts. The five-run mean in Table 1 is a different estimate. Within the human subset, dual–single differences are below 0.03 and vary in direction. The visible negative-transfer loss is therefore concentrated in the AI-side quad-

Model	Full set	Chinese	English	Human subset	Human-zh	Human-en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head	0.663 [.646, .680]	0.819 [.802, .836]	0.429 [.399, .458]	0.629 [.605, .653]	0.774 [.748, .798]	0.399 [.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 12: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for RCNN).

rants, consistent with §5.3.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI $\times$ non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore shifts across the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

AUC values within AI texts are similar across models (0.82–0.88). Joint training thus raises sarcasm scores while largely preserving their internal ordering. This pattern is consistent with a source-related component being added to the sarcasm signal.

D Representative Cases of Detector–LLM Score Misalignment

Cases were manually selected after inspecting the paired table of RCNN-detector scores and ratings from the three LLMs to illustrate large disagreements; no formal ranking or top- k selection was implemented. Detector scores range from 0 to 1, and LLM scores are the three-evaluator mean on a 1–5 scale; long texts are truncated. Chinese examples retain the original model input and provide a pinyin transliteration and an English translation in brackets.

D.1 Classified as human-authored by the detector but rated as less human-like by the LLMs (MGTD-score misalignment; corresponding to the analysis in §6). In the first case, AI-generated text with obvious chain-of-thought traces is nevertheless assigned human-authorship probability 1.00 by the MGTD head, corresponding to the score disagreement discussed in §6.

D.2 Detector-high / LLM-judge-low sarcasm cases.

D.3 Detector-low / LLM-judge-high sarcasm cases. Together, these selected cases motivate the hypothesis that surface markers such as mock-praise formulas, exclamation marks, and collo-

quial short sentences contribute to some detector decisions, while semantic reversal contributes to some LLM-judge ratings. Systematic feature-level testing is required to determine whether either pattern generalizes.

The manually selected cases do not estimate how often either pattern occurs. They provide qualitative examples of the near-zero Chinese sarcasm correlation in §6; systematic testing remains future work.

L	S	$P_{\text{hum}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [hǎo, wǒ xūyào zhēnduì Sānxīng huíduì Huáwéi de xīnwén shēngchéng yī tiáo fěngcì pínglùn. shǒuxiān...] [“Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First...”] (chain-of-thought leakage)
zh	SFT	1.00 / 2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [pǎofēn jìngsài: cānshù jiāolǜ: pǎofēn jiāolǜ: shuǐhuò: hāhāhā] [“Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
en	SFT	1.00 / 2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 13: MGTD-score misalignment cases. $P_{\text{hum}} = P(\text{human})$.

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [Sānxīng: wǒ cái shì dì-yī! nǐ kànkān wǒ zhè shèxiàngtóu zǒngfēn, 109+96, 112+89, wǒ yíng le!] [“Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
zh	SFT	0.99 / 2.67	“恭喜恭喜，美国牛牛牛，继续加油” [gōngxǐ gōngxǐ, Měiguó niú niú niú, jìxù jiāyóu] [“Congratulations, congratulations, America is awesome awesome awesome, keep it up”] (formulaic mock-praise)
en	SFT	0.93 / 2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 14: Detector-high / LLM-judge-low sarcasm cases. $P_{\text{sar}} = P(\text{sarcastic})$.

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	Base	0.14 / 4.00	“特朗普承诺 120 亿补贴像撒大饼，农民收到的却是缩水成硬币的承诺……” [Tèlǎngpǔ chéngnuò 120 yì bǔtiē xiàng sā dàbǐng, nóngmín shōudào de què shì suōshuǐ chéng yìngbì de chéngnuò...] [“Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins...”] (metaphorical irony)
zh	Base	0.24 / 4.00	“特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [Tèsīlā: jiàgé tiàoshuǐ hòu yòu tiàoshuǐ, xiāofèizhě: qiānbāo tiàoshuǐ hòu yòu tiàoshuǐ.] [“Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
en	Base	0.12 / 4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis)

Table 15: Detector-low / LLM-judge-high sarcasm cases. $P_{\text{sar}} = P(\text{sarcastic})$.

E Preliminary Representation-Level Analysis of the Failure Mechanism

This appendix records the complete protocol, data, and self-checks for the representation-level tests in §5.3.2. The interpretation of conclusions follows §5.3.2.

E.1 Setup. All analyses freeze the plain dual-head model that produces the 86.3% main-text result. We use split-sample evaluation: one half estimates the direction and the other computes metrics. The appendix baseline is therefore 84.8% on the evaluation half rather than 86.3% on the full set.

Implementation checks pass. Manually computed logits match model outputs with maximum error 1.4×10^{-6} , and the erasure-projection residual is about 10^{-6} . LEACE satisfies its defining property: after erasure the fitting-half class-mean difference has relative residual 7×10^{-6} , and an in-sample linear probe is at chance (AUC 0.514). Probe selectivity is 0.986 with true labels and 0.512 with permuted labels.

E.2 Protocol. The linear probe is logistic regression with five-fold cross-validation. A selectivity control retrains it with permuted source labels, and correlation analyses are repeated within text-length tertiles. For alignment, Σ is the Ledoit–Wolf covariance fitted to all pooled Dual-Test representations, with eigenvalues clipped below 10^{-8} ; w denotes a task-head weight differ-

ence or probe direction. Whitened alignment is $\cos(\Sigma^{1/2}w_1, \Sigma^{1/2}w_2)$.

Erasure tests cover one direction, INLP subspaces ($k = 1-16$; Ravfogel et al., 2020), and LEACE (Belrose et al., 2023). Controls use ten sampled random whitened directions and directions estimated from permuted labels. srcORTH-PC1 is the maximum-variance direction after removing its projection onto the estimated linear source direction. Erasing it leaves measured linear source decodability unchanged, although orthogonality to one estimated direction does not imply independence from all multidimensional or nonlinear source information.

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts. The permuted-label control gives 0.512. The single-head model also gives 0.974, so source decodability is not unique to joint training. On AI $\times$ non-sarcastic texts, the MGTD-head AI logit and sarcasm-head logit correlate at $\rho = 0.519$; correlations by text-length tertile are 0.44, 0.56, and 0.63.

The sarcasm and source directions are nearly orthogonal in the original geometry ($\cos \approx -0.03$) but moderately aligned after whitening. The absolute cosine is 0.41 using the probe direction and 0.49 using the MGTD-head direction. These results are compatible with the shortcut hypothesis, but remain correlational.

E.4 Intervention evidence. $\text{gap} = \text{AI} \times \text{non-sarcastic false-positive rate} - \text{human} \times \text{non-sarcastic false-positive rate}$; source AUC is the decoding ability of a probe retrained after erasure.

Erasure scheme	AI $\times$ n FP	H $\times$ n FP	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
Single dir. (head / probe / random)	84.8–86.2	—	—	≈ 0.98
INLP ($k = 16$)	89.8	—	—	0.974
LEACE (linear)	68.7	49.5	19.1	0.589
Random	84.9	32.7	—	0.982
whitened $\times 10$				
srcORTH-PC1	66.7	54.1	12.6	0.982

Table 16: Intervention evidence. H $\times$ n = human $\times$ non-sarcastic; AI $\times$ n = AI $\times$ non-sarcastic; FP in %.

Single-direction and low-dimensional INLP erasure produce little change. LEACE reduces measured linear source decodability and the source-specific gap from 52.0 to 19.1; the ten sampled random-direction controls show no comparable

gap change.

The orthogonal high-variance control changes the interpretation. It leaves measured linear source decodability intact but changes the AI-side rate to 66.7 and the source gap to 12.6, an effect similar to LEACE. This shows that predictions are sensitive to at least one high-variance direction after removing its projection onto the estimated source direction; it does not identify an independent or unique causal subspace.

Under LEACE, the unweighted mean false-positive rate across the two non-sarcastic source groups is almost unchanged ($58.8 \rightarrow 59.1$). The intervention therefore reduces the source disparity primarily by shifting errors from AI texts to human texts rather than improving non-sarcastic classification overall.